BAIS 4720

Netflix Dashboard Project Report

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1. Data Handling

This report will present an analysis of Netflix content performance using datasets obtained from the FlixPatrol API. In this analysis, I examined viewership patterns and content type distribution for the 1st quarter of 2023, as the data allowed.

There was a wide range of data included in the API, including social popularity, most-watched content, market data, and country data for various streaming platforms. This analysis focuses mostly on Netflix. Software used: Visual Studio Code, Bruno API client, Microsoft Excel, and Power BI.

The only monetary expense this project required was the API key. Once it was paid for, I authenticated the API key using the Bruno extension on VSCode. The Bruno extension is used to test APIs and was a recommendation from the FlixPatrol website. Using Bruno, I extracted the data into nested JSON files before converting them into CSV format. This was one of the most challenging parts of this project. After this, I utilized Microsoft Excel to clean and merge the datasets. After cleaning, I imported the Excel files into Power BI to create the Netflix Performance Dashboard.

The raw files and cleaned data sets will be turned in with this report.

2. Research questions

a. Which TV show and movie titles have the most viewing hours?
b. What is the market share for both content types?
c. What is the relationship between the number of hours watched and the number of total views?

The answers to these questions are illustrated in the dashboard.

3. Findings

From the most_watched dataset, TV shows make up 55.18% of total viewership (about 3 billion total views), while movies take up the remaining 44.82%. Although undisclosed, this includes all available genres within the datasets, e.g., romance, action, and documentaries.

In the scatter plot, the relationship between movie and TV show viewership is apparent. Since TV shows have longer runtimes than movies, they tend to have a higher number of total hours watched, along with lower views compared to movies. And subsequently, movies have shorter runtimes, so they can often have more views. The titles in the top right corner of the scatter plot (high views and high total hours) are very successful, usually due to quality and re-watchability.

4. Lessons Learned

- The main thing I learned is how to access, authenticate, and extract data from an API key
- Easier and faster ways to clean and merge datasets
- The difference between JSON and CSV files and how they can be used

5. Challenges

- The learning curve associated with API data
- Lack of relevant data
- Having to pay for access to accurate information